Project Report

Multi Agent Insurance System

An AI-Powered Multi-Agent Platform for Insurance Policy Research, Comparison & Explanation

Course Name: Agentic AI

Institution Name: Medicaps University

Datagami Skill Based Course

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1. Summary

1. The Multi-Agent Insurance System project presents the design and implementation of an AI-driven collaborative platform that automates insurance policy research, comparison, and explanation using specialized intelligent agents. The system integrates multiple agents including Researcher Agent, Comparator Agent, Explainer Agent, and Coordinator Agent to perform structured insurance analysis and generate user-friendly outputs.

2. The platform leverages a multi-agent orchestration framework with Google Gemini LLM integration to process user queries, retrieve insurance policy information, compare plans based on predefined criteria, and provide simplified explanations. The agents operate collaboratively under a coordinator workflow, ensuring modular execution and accurate decision support.

3. Key capabilities of the system include automated policy research, intelligent policy comparison, simplified explanation of policy details, and structured response generation. The modular architecture enhances scalability, extensibility, and maintainability while reducing manual effort in insurance analysis tasks.

4. The system addresses common challenges in insurance decision-making such as difficulty understanding policies, comparing multiple plans, and interpreting coverage details. The solution demonstrates strong potential for improving customer understanding, accelerating insurance selection, and enabling AI-assisted policy evaluation.

2. Introduction

The insurance industry involves complex processes such as selecting suitable policies, comparing coverage benefits, evaluating risk factors, and understanding policy documents. Traditional insurance systems rely heavily on manual analysis and human intermediaries, which often results in slower decision-making, limited transparency, and increased operational workload.

To address these challenges, this project proposes a Multi-Agent Insurance System that uses collaborative AI agents to automate policy research, comparison, and explanation. The system allows users to request policy information, compare multiple insurance plans, and receive simplified explanations through an intelligent agent-driven workflow.

By integrating multi-agent architecture with Large Language Model (LLM) capabilities, the system provides structured and accurate insurance insights. This approach enhances decision-making efficiency, reduces manual workload, and improves accessibility of insurance information for users.

2.1Background of the Study

3. Insurance policy selection requires analyzing multiple parameters such as coverage, premium, eligibility conditions, benefits, and exclusions. Customers often rely on manual research or agents to interpret policy details, which can be time-consuming and prone to misunderstanding due to complex terminology.
4. Advancements in Artificial Intelligence and multi-agent systems enable automation of such analytical tasks by distributing responsibilities among specialized agents. The adoption of LLMs further enables natural language understanding and simplified explanation of technical policy information.
5. The Multi-Agent Insurance System leverages these technologies to automate policy research and comparison, improving efficiency and accuracy in insurance decision support.

2.2Overview of Agentic AI in Insurance

Agentic AI refers to autonomous systems where multiple specialized agents collaborate to achieve complex objectives. In the insurance domain, such systems enable intelligent automation of policy analysis and recommendation workflows.

The proposed system consists of specialized agents including Researcher Agent, Comparator Agent, Explainer Agent, and Coordinator Agent. Each agent performs a dedicated function while communicating through the coordinator to complete end-to-end insurance workflows.

The architecture diagram in the HLD shows how user requests are processed by the coordinator and delegated to agents for research, comparison, and explanation before generating final output.

2.3Importance of LLM-Based Policy Understanding

Insurance policies contain complex legal and technical language that is difficult for users to interpret. Integrating Google Gemini LLM enables the system to process policy content and generate simplified explanations in natural language.

LLM-based reasoning provides:

- Natural language interaction
- Simplified policy descriptions
- Context-aware explanations
- Structured outputs

This improves user understanding and confidence in insurance decisions.

2.4 Industry Context: AI in Insurance

The insurance industry is increasingly adopting AI technologies for policy recommendation, risk evaluation, and customer assistance. However, many existing solutions rely on monolithic systems with limited automation and user-friendly explanation capabilities.

The Multi-Agent Insurance System addresses these limitations by enabling modular AI agents that independently perform policy research, comparison, and explanation while collaborating through orchestration. This approach supports scalable automation and improved customer experience in digital insurance platforms.

3.Problem Statement & Objectives

3.1 Problem Statement

Customers face several challenges when selecting and comparing insurance policies:

- Complex policy documents
- Difficulty comparing multiple plans
- Limited understanding of coverage and exclusions
- Manual research requirements
- Dependence on human intermediaries

Traditional systems do not provide automated comparison or simplified explanations of policies. There is a need for an AI-driven system capable of researching, comparing, and explaining insurance policies autonomously.

3.2 Project Objectives

The main objective is to develop a Multi-Agent Insurance System capable of:

1. Understanding user policy queries
2. Researching insurance policy details automatically
3. Comparing multiple insurance plans
4. Generating simplified explanations
5. Coordinating multi-agent workflow
6. Providing structured decision support

3.3 Scope of the Project

Included

- Insurance policy research automation
- Policy comparison functionality
- Simplified explanation generation
- Multi-agent orchestration architecture
- LLM integration

Out of Scope

- Real-time insurer integration
- Payment processing
- Full claim settlement
- Regulatory compliance automation

4. Proposed Solution

The proposed solution is a Multi-Agent Insurance System designed to automate policy research, comparison, and explanation using collaborative AI agents coordinated by a central orchestrator.

The system operates similarly to an insurance advisory team where each agent performs a specialized task and contributes to the final decision output.

4.1 Key Features

- Multi-agent collaborative architecture
- Automated policy research
- Intelligent policy comparison
- Simplified policy explanation
- LLM-based reasoning
- Modular and scalable design

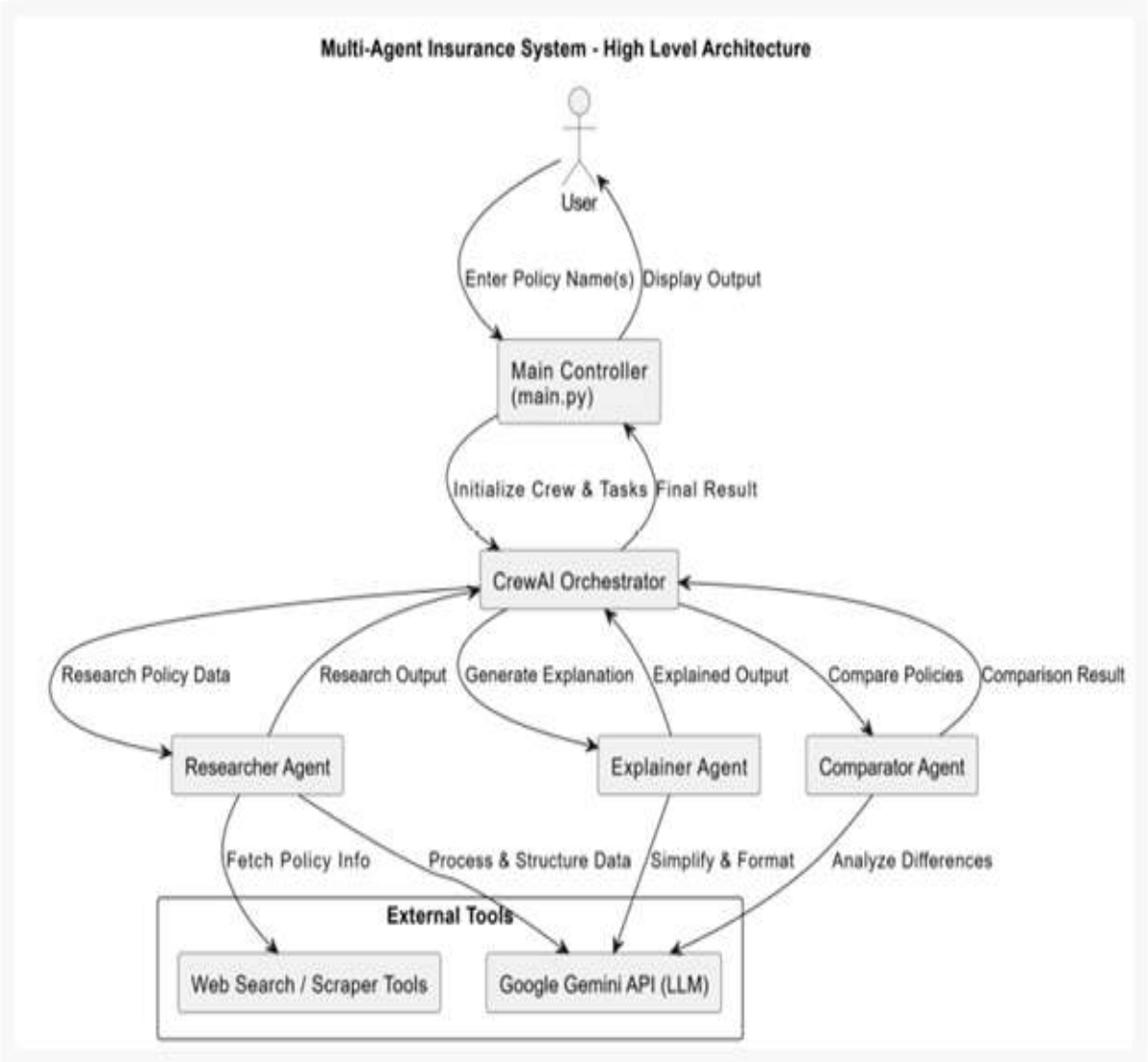
4.2 Overall Architecture / System Workflow

The The workflow follows these steps:

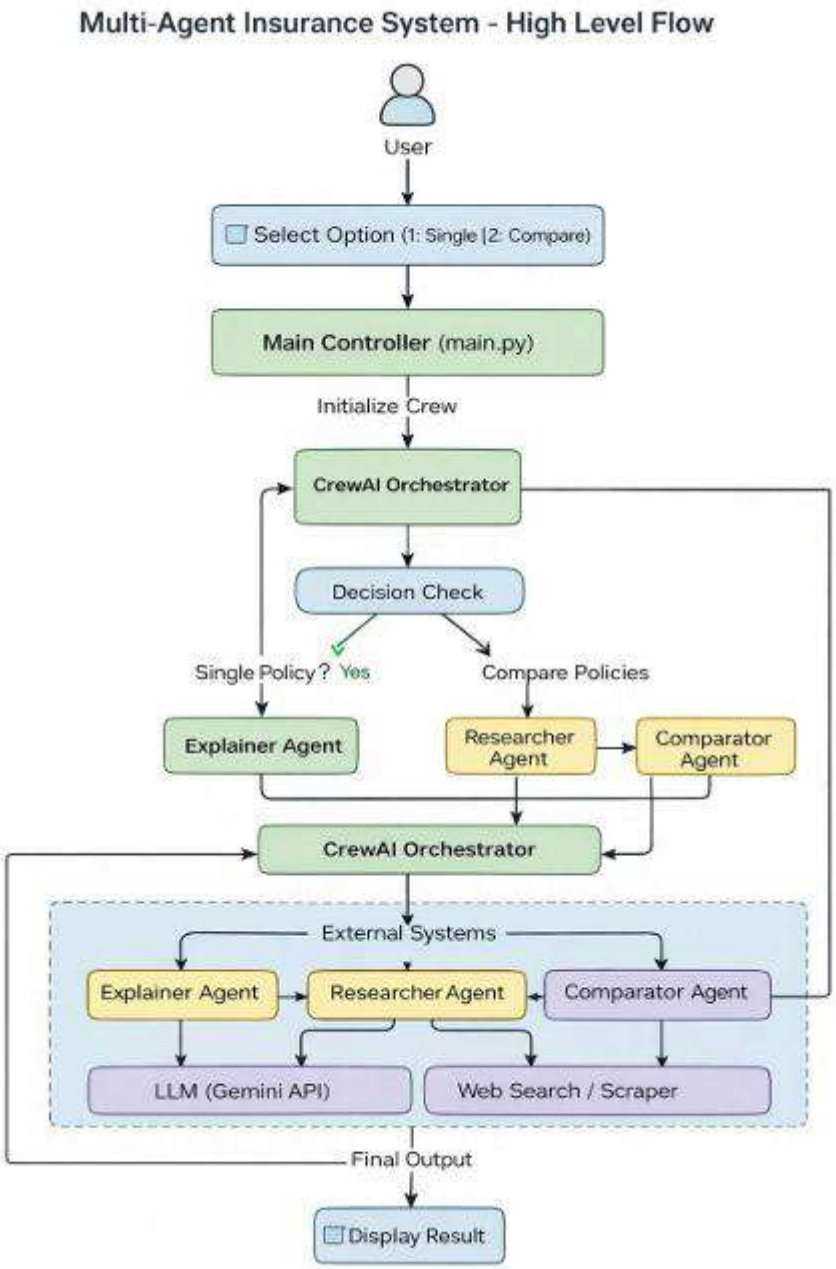
1. User submits policy query
2. Coordinator Agent receives request
3. Request routed to Researcher Agent
4. Comparator Agent evaluates policies

5. Explainer Agent generates explanation
6. Final output returned to use

4.2.1 High-Level Architecture Diagram



4.2.2 System Workflow Flowchart



4.2.3 Agent Processing Pipeline

4.3 Agent Design & Role Definition

4.3.1 Coordinator Agent

Coordinates workflow, receives user requests, delegates tasks to agents, and aggregates outputs.

4.3.2 Researcher Agent

Collects insurance policy information including coverage, premium, benefits, and exclusions.

4.3.3 Comparator Agent

Compares policies based on cost, coverage, and suitability criteria and ranks options.

4.4 LLM Integration Strategy

Google Gemini LLM is integrated to process policy data and generate simplified explanations. The LLM combines research data and comparison results to produce structured outputs that are easy for users to understand.4.4 RAG Implementation Strategy

4.5 System Design Decisions

- Multi-agent architecture for modularity
- Orchestrator-based workflow
- REST APIs for communication
- LLM integration for explanation
- Independent agent modules for scalability

4.6 Tools & Technologies Used

Component	Technology	Purpose
Backend	Python	Core logic
AI Framework	CrewAI	Agent orchestration
LLM	Google Gemini	Explanation
API	REST APIs	Communication
Database	Structured DB	Policy data

5.System Implementation

The Multi-Agent Insurance System is implemented as a modular collaborative architecture in which specialized agents execute insurance policy analysis tasks under the supervision of a Coordinator Agent. The implementation integrates a Python-based backend, CrewAI multi-agent orchestration, Google Gemini LLM for natural language reasoning, and structured policy data storage.

The system processes user insurance queries, retrieves relevant policy information, compares multiple plans, and generates simplified explanations through a controlled multi-agent workflow. Each module operates independently while communicating through the orchestrator to ensure consistent and reliable outputs

5.1 Query Routing Mechanism

The query routing mechanism is responsible for receiving user inputs and directing them to the appropriate agents based on the requested insurance task. User queries are submitted through the interface and passed to the Coordinator Agent, which acts as the central decision controller.

The Coordinator Agent performs initial analysis of the request to determine required processing stages. For example, a policy comparison request triggers the Researcher Agent followed by the Comparator Agent and Explainer Agent. The routing logic ensures that agents execute in the correct sequence while maintaining workflow integrity.

The routing mechanism also maintains task logs and state information to track execution progress. This controlled routing approach prevents redundant agent execution and ensures that each request follows the predefined multi-agent pipeline.

5.2 Dataset Processing Module

The Policy Data Processing Module is implemented within the Researcher Agent. This module retrieves insurance policy information from structured storage or external sources and converts it into standardized data suitable for comparison and explanation.

The module extracts key policy attributes including coverage details, premium values, eligibility conditions, benefits, and exclusions. Data normalization techniques are applied to ensure consistent formatting across policies, enabling accurate comparison. Validation checks are performed to ensure completeness and correctness of retrieved information before passing it to downstream agents.

The processed policy dataset serves as the primary input for the Comparator Agent. By structuring policy information into a uniform representation, the system enables efficient analysis and reliable decision support.

5.3 Agent Collaboration Workflow

The Multi-Agent Insurance System follows a sequential collaboration workflow where each agent processes validated input from the previous stage. This ensures controlled hand-off and consistent results.

The workflow begins with the Researcher Agent collecting and structuring policy data. The Comparator Agent then evaluates multiple policies using predefined criteria such as cost

efficiency, coverage value, and suitability. The results are passed to the Explainer Agent, which generates simplified natural-language descriptions and final structured outputs.

The Coordinator Agent manages communication between agents, ensuring that each stage executes only after completion of prerequisite tasks. This controlled pipeline prevents incomplete processing and maintains workflow integrity.

5.4 LLM Integration & Response Generation

The Explainer Agent integrates Google Gemini LLM to convert structured insurance data into user-friendly explanations. The LLM receives combined inputs from the Researcher and Comparator agents, including policy attributes and comparison results.

Using natural language reasoning, the LLM generates simplified descriptions of policy coverage, benefits, and differences between plans. The generated response is formatted into structured output containing policy summary, comparison insights, and recommendation context.

This LLM-based response generation ensures clarity, readability, and accessibility of insurance information for users. It also enables consistent formatting across outputs regardless of input complexity.

5.5 State and Workflow Management

Google The system maintains workflow state across agent execution stages to ensure controlled progression. Each request transitions through defined states such as request received, data collected, policies compared, and response generated.

The Coordinator Agent validates state transitions before activating subsequent agents. If an error occurs, the workflow remains at the last stable state, preventing inconsistent outputs. Logging mechanisms track state transitions for monitoring and debugging.

This structured state management ensures reliable multi-agent collaboration and prevents execution errors during complex insurance processing tasks.

5.6 Implementation Architecture Integration

The implemented architecture integrates multiple layers including user interface, orchestrator layer, agent layer, LLM services, and policy data storage. The interface captures user requests

and forwards them to the Coordinator Agent through REST-based communication. Agents operate within the backend environment and interact with structured policy data sources.

Google Gemini LLM services are invoked by the Explainer Agent to generate explanations. The modular architecture allows independent scaling or modification of agents without affecting other components. This layered integration supports extensibility and maintainability of the insurance automation platform.

6. Results & Output

The Multi-Agent Insurance System was evaluated using multiple insurance policy queries involving research, comparison, and explanation tasks. The results demonstrate the system's capability to autonomously retrieve policy information, compare plans based on structured criteria, and generate simplified explanations with minimal human intervention.

The collaborative operation of the Researcher, Comparator, and Explainer agents successfully produced accurate and structured outputs across different insurance scenarios.

6.1 Policy Research Output

The Researcher Agent successfully retrieved and structured insurance policy data including coverage details, premium values, eligibility conditions, benefits, and exclusions. The agent converted unstructured policy information into standardized structured format suitable for downstream comparison and explanation.

Example Output Observations:

- Accurate extraction of policy attributes
- Consistent data structuring across policies
- Validation of policy completeness

This demonstrates the effectiveness of the research module in automating policy information gathering without manual intervention

6.2 Policy Comparison Output

Comparator Agent evaluated multiple policies using predefined criteria such as cost efficiency, coverage value, and suitability. The system successfully identified key differences between policies and ranked options based on comparative metrics.

Example Comparison Insights:

- Premium vs coverage trade-off

- Benefit differences
- Suitability ranking
- Policy strengths and weaknesses

The comparison module enabled automated multi-policy evaluation that traditionally requires significant manual effort.

6.3 Generated Explanation Output

The Explainer Agent integrated Google Gemini LLM to convert structured policy and comparison data into simplified natural-language explanations. The system produced user-friendly summaries describing coverage, benefits, and differences between policies.

Example Explanation Features:

- Simplified policy descriptions
- Easy-to-understand comparison insights
- Clear recommendation context
- Structured response formatting

This demonstrates the effectiveness of LLM integration in improving accessibility of insurance information for non-expert users.

6.4 System Performance Observations

The system demonstrated stable and consistent performance across multiple query scenarios. Agent collaboration executed sequentially without workflow conflicts, and response generation remained consistent regardless of policy complexity.

Observed Advantages:

- Automated multi-agent execution
- Reduced manual analysis effort
- Consistent structured outputs
- Fast response generation
- Improved decision clarity

The results confirm that the Multi-Agent Insurance System can effectively automate insurance policy research, comparison, and explanation workflows.

6.5 Screenshots/System outputs

7. Limitations

While the Multi-Agent Insurance System demonstrates effective automation of insurance policy analysis, the current implementation has certain limitations that may affect scalability and real-world deployment.

7.1 Limited Real-Time Policy Integration

The system currently relies on predefined or locally available policy data. Real-time integration with insurance provider databases or APIs is not implemented. As a result, policy updates or new offerings may not automatically reflect in the system.

7.2 Dependence on Data Quality

Accuracy of research, comparison, and explanation outputs depends on the completeness and correctness of policy data. Missing or inconsistent information can affect comparison accuracy and explanation clarity.

7.3 Limited Claim Processing Capability

The current system focuses primarily on policy research and comparison. Full claim lifecycle automation, including validation, fraud detection, and settlement workflows, is not implemented.

7.4 LLM Interpretation Variability

Although Gemini LLM improves explanation quality, natural language generation may occasionally vary in phrasing or emphasis. Explanation accuracy depends on structured input quality and prompt design.

7.5 Scalability Constraints

The system has been evaluated on limited policy datasets. Deployment across large-scale insurance catalogs or enterprise environments may require optimization of data handling and agent orchestration.

Despite these limitations, the system provides a strong foundation for AI-driven insurance decision support and multi-agent automation.

8. Conclusion

This project presented the design and implementation of a Multi-Agent Insurance System that automates insurance policy research, comparison, and explanation using collaborative AI

agents. The system integrates specialized agents including Researcher, Comparator, and Explainer agents under a Coordinator-based orchestration architecture.

The proposed solution successfully demonstrates the following capabilities:

- Automated retrieval of insurance policy information
- Intelligent comparison of multiple policies
- Simplified natural-language explanation generation
- Structured multi-agent collaboration workflow
- LLM-based insurance decision support

The modular multi-agent architecture ensures scalability, extensibility, and clear separation of responsibilities between agents. Integration of Google Gemini LLM enhances readability and accessibility of complex policy information, improving user understanding and decision confidence.

Overall, the Multi-Agent Insurance System provides a practical framework for applying Agentic AI in insurance policy analysis and advisory workflows. The system reduces manual effort, accelerates comparison processes, and improves transparency in insurance decision-making.

9. Future Scope & Enhancements

The current implementation establishes a foundational AI-driven insurance analysis platform. Several enhancements can extend the system into a comprehensive intelligent insurance assistant.

9.1 Real-Time Insurance Provider Integration

Integration with live insurance provider APIs would enable automatic retrieval of updated policy data, premium changes, and new offerings. This would improve accuracy and real-world applicability.

9.2 Automated Claim Processing Agent

Future versions can include a Claim Processing Agent capable of validating claims, checking policy eligibility, and automating approval workflows. This would extend the system beyond policy analysis into full insurance lifecycle automation.

9.3 Personalized Policy Recommendation

Integration of user profiling and risk assessment models can enable personalized policy recommendations tailored to customer demographics, financial status, and coverage needs.

9.4 Fraud Detection Module

AI-based fraud detection agents can analyze claim patterns and policy usage to identify anomalies or suspicious activities, improving insurance risk management.

9.5 Web-Based Dashboard and Visualization

A graphical dashboard can provide interactive policy comparison charts, coverage visualization, and recommendation insights to improve user interaction and decision clarity.

9.6 Multi-Language Explanation Support

LLM-based multilingual explanation capability can make insurance policies understandable across diverse linguistic users, improving accessibility.

These enhancements would transform the system into a fully intelligent insurance advisory and automation platform suitable for enterprise deployment.

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